

Gabriel Cruz

240-810-4092 | gabriel.david.cruz12@gmail.com | [linkedin.com/in/gcruz38](https://www.linkedin.com/in/gcruz38) | github.com/gabrieldcruz

EDUCATION

Georgia Institute of Technology

B.S. in Computer Science and Mathematics, GPA: 3.2

Atlanta, GA

Aug 2024 - May 2028

- Threads: Systems & Architecture, Intelligence
- Coursework: Data Structures & Algorithms, Design & Analysis of Algorithms, Computer Architecture, Systems & Networks, Discrete Mathematics, Linear Algebra, Probability & Statistics

PROJECTS

Wisp: Privacy-Friendly Analytics | Java, Spring Boot, Redis, ClickHouse, React

- Built a cookie-free analytics platform: a 947-byte tracker, a Spring Boot ingest API making one atomic Redis Lua call per event (rate limits, stream buffering, HyperLogLog uniques, live visitors), and a React dashboard
- Designed exactly-once rollups over at-least-once Redis Streams delivery using batch deduplication tokens in ClickHouse, verified with crash-injection integration tests
- Batched ClickHouse inserts for 640× the throughput of single-row writes; materialized-view rollups cut dashboard queries 2-22×; 49 unit and Testcontainers integration tests run in CI

Cosmillion: Multiplayer Trivia Game | Django, PostgreSQL, SvelteKit, Docker, AWS

- Built a full-stack game with a PostgreSQL schema (13 migrations, constraints, ordered row locks), party rooms with synchronized timed rounds, and a daily challenge with streaks and a leaderboard
- Generated a reproducible bank of 703 questions and ~39,000 answers from Wikidata and Wikipedia pageviews
- Wrote 76 backend tests and 2 Playwright browser tests; GitHub Actions CI auto-deploys passing merges to EC2 (Docker, CloudFront)
- Load-tested the live app, traced a CPU bottleneck to per-request database connections, and fixed it for +69% throughput and 42% lower median latency

Personal Portfolio Website | SvelteKit, TypeScript, CSS, Vercel

- Built a responsive six-page site with typed TypeScript models powering interactive case studies for 10 projects, tabbed content, and image/video dialogs
- Added keyboard controls, modal focus restoration, and reduced-motion support; persisted audio and motion preferences with Svelte context and localStorage
- Built a Web Audio API module that synthesizes navigation sounds with oscillators and gain envelopes, with user-gesture activation, event throttling, and cleanup

Multithreaded CPU Scheduler | C, pthreads, Linux

- Implemented FCFS, Round Robin, SRTF, and preemptive priority schedulers for a multiprocessor OS simulator with one thread per simulated CPU
- Protected shared ready queues and per-CPU state with mutexes and condition variables, synchronizing access and blocking idle CPUs

EXPERIENCE

Electrical Engineering Intern

June 2023 - May 2024

Johns Hopkins Applied Physics Laboratory

Laurel, MD

- Led small teams building prototypes for wireless power transfer (WPT) and electrical impedance spectroscopy (EIS), including a low-cost 2.45 GHz closed-loop WPT system with adaptive phase and impedance tuning
- Simulated 2.45 GHz and 5.8 GHz ISM signals in GNU Radio, reducing energy rebound by 30%; developed a predictive Free Space Path Loss model to improve wireless system design accuracy
- Built an automated leaf health monitoring pipeline using an Arduino and Python (Plotly) that counted xylem in leaves and flagged whether they fell in a healthy range, achieving 99.1% detection accuracy

TECHNICAL SKILLS

Languages: Java, Python, C, C++, TypeScript, JavaScript, SQL, Assembly, MATLAB

Backend & Frontend: Spring Boot, Django REST Framework, React, SvelteKit, REST APIs

Data & Infrastructure: PostgreSQL, MySQL, ClickHouse, Redis, Docker, AWS, GitHub Actions

Testing & Tools: JUnit, Testcontainers, pytest, Playwright, Linux, Git, GDB, Make, pthreads